

David Cornelis Groothuizen Dijkema

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Education

University of Auckland | Waipapa Taumata Rau

Doctoral Candidate in Mathematics Mar. 2022 – present

Thesis topic: Dynamical dependence on topology of symmetric heteroclinic networks

Supervisors: Profs. Claire Postlethwaite and Vivien Kirk

Bachelor of Science (Honours) 2021

with First Class Honours in Applied Mathematics

Honours Dissertation: A heteroclinic network in a spatially-extended four species interaction model

Supervisor: Prof. Claire Postlethwaite

Bachelor of Engineering (Honours) 2017 – 2020

with First Class Honours in Engineering Science

Part IV Project: Synthetic-aperture radar processing on the NVidia Jetson AGX Xavier

Supervisors: Prof. John Cater and Assoc Prof. Nicholas Rattenbury

Publications and preprints

David C Groothuizen Dijkema and Claire M Postlethwaite. (2023). Travelling waves and heteroclinic networks in models of spatially-extended cycle competition. *Nonlinearity*, **36**(12) 6546–6588.

DOI: [10.1088/1361-6544/ad0212](https://doi.org/10.1088/1361-6544/ad0212). arXiv: [2208.05630](https://arxiv.org/abs/2208.05630) [math.DS].

David C Groothuizen Dijkema, Vivien Kirk, and Claire M Postlethwaite. Analysis of dynamics near heteroclinic networks in $\mathbb{R}^4$ with a projected map. *Submitted to SIAM Journal on Applied Dynamical Systems*.

arXiv: [2410.21486](https://arxiv.org/abs/2410.21486) [math.DS].

David C Groothuizen Dijkema, Vivien Kirk, Claire M Postlethwaite, and Alastair M Rucklidge. Continuity of projected maps near heteroclinic networks in $\mathbb{R}^4$. *In preparation. Preprint available upon request.*

Awards and scholarships

Nonlinearity Outstanding Reviewer Award 2023

University of Auckland Academic Career Advancement Award

Australia and New Zealand Industrial and Applied Mathematics Prize
for Best Poster Presentation

Honourable mention for the New Zealand Mathematical Society Aitken Prize
for Best Student Talk

Marsden Fund Doctoral Scholarship 2022

University of Auckland Department of Mathematics Student Research Conference prize 2021

First in Course Award in MATHS 761 (Dynamical Systems)
for MATHS 761 (Dynamical Systems)

University of Auckland Postgraduate Honours Scholarship	
University of Auckland Summer Research Scholarship	2020
University of Auckland Faculty of Engineering Dean's Honours List	2017, 2018, 2019
First in Course Award	2019
for ENGSCI 355 (Applied Modelling in Simulation and Optimisation)	

Teaching experience

Department of Mathematics, Faculty of Science, University of Auckland	2022 – 2024
Academic Casual Staff	
Wrote teaching material, MATLAB templates, and completed MATLAB examples used to teach CoCo in MATHS 761, a postgraduate course on dynamical systems theory.	
Graduate Teaching Assistant for	
COMPSCI 120 (Mathematics for Computer Science)	
COMPSCI 225 (Discrete Structures in Mathematics and Computer Science)	
MATHS 108 (General Mathematics 1)	
MATHS 120 (Algebra 1)	
MATHS 130 (Calculus 1)	
MATHS 253 (Algebra and Calculus 3)	
MATHS 260 (Differential Equations)	
Mathematics Assistance Room	
Department of Engineering Science, Faculty of Engineering, University of Auckland	2020
Teaching Assistant for	
ENGSCI 255 (Modelling and Analytics in Operations Research)	
ENGSCI 211 (Mathematical Modelling 2)	

Professional development

Academic Career Advancement Programme (ACAP)	2024
A year-long course run by the School of Graduate Studies for 20 doctoral candidates, featuring 13 three-hour sessions covering different aspects of an academic career. Ten of these sessions are organised by ACAP participants in pairs. I co-organised with Nova Ariani a seminar on the <i>Broader Dissemination of Research</i> , which included a one-hour workshop with Prof. Neal Curtis on using ideas from comics to design posters, and a talk by Prof. Cliona Ni Nhurchú on engaging with communities, industry, and government.	
Department of Mathematics Tutor Training	2022
A three-hour workshop run by the Department, focussing on the fundamentals of interacting with students in tutorials, being an engaging and effective tutor, and cultural sensitivity.	

Research visits

School of Mathematics, University of Leeds, UK	Jun. 5 – Jul. 7, 2023
Visited Prof. Alastair Rucklidge to work on understanding the continuity of projected maps of heteroclinic networks in $\mathbb{R}^4$.	

Centre of Mathematics, University of Porto, Portugal

Jun. 19 – 23, 2023

Visited Prof. Sofia Castro to work on various problems concerning the dynamics of trajectories near heteroclinic networks in $\mathbb{R}^4$.

Presentations

Dynamics near heteroclinic networks. <i>Applied Mathematics Unit Doctoral Talks.</i> University of Auckland, NZ.	Oct. 2024
Switching and cycling near heteroclinic networks as a piecewise smooth dynamical system. <i>Australia New Zealand Applied Mathematics Conference.</i> Hahndorf, SA, Australia.	Feb. 2024
Switching and cycling near heteroclinic networks as a piecewise smooth dynamical system. <i>New Zealand Mathematical Society Colloquium.</i> Victoria University of Wellington, NZ.	Dec. 2023
Travelling waves and heteroclinic networks in spatially-extended models of cyclic competition. <i>New Zealand Mathematics and Statistics Postgraduate Conference.</i> Wellington, NZ.	Nov. 2023
Dynamics near heteroclinic cycles and networks. <i>Leeds Applied Nonlinear Dynamics Seminar.</i> University of Leeds, UK.	Jun. 2023
Dynamics near heteroclinic cycles and networks. <i>Centro de Matemática da Universidade do Porto Seminar.</i> University of Porto, Portugal.	Jun. 2023
Switching near heteroclinic networks as a piecewise-smooth dynamical system. <i>Minisymposium on Long-Term Dynamics, SIAM Conference on Applications of Dynamical Systems.</i> Portland, OR, USA.	May. 2023
Switching near heteroclinic networks as a piecewise-smooth system. <i>Sydney Dynamics Group Workshop: Dynamics in NZ.</i> Karioitahi, NZ.	Nov. 2022
Switching near heteroclinic networks as a piecewise-smooth system. <i>Applied Mathematics Unit Doctoral Talks.</i> University of Auckland, NZ.	Oct. 2022
Stability and cycling near heteroclinic cycles and networks. <i>Department of Mathematics Student Research Conference.</i> University of Auckland, NZ.	Jun. 2022
Stability and cycling near heteroclinic cycles and networks. <i>Department of Mathematics Postgraduate Seminar.</i> University of Auckland, NZ.	Jun. 2022
Spatially-extended models of competition between four species. <i>Department of Mathematics Postgraduate Mid-year Talks.</i> University of Auckland, NZ.	Jul. 2021
A spatially-extended model of competition between four species. <i>Department of Mathematics Student Research Conference.</i> University of Auckland, NZ.	Jun. 2021
Heteroclinic networks in partial differential equations. <i>Department of Mathematics Summer Research Conference.</i> University of Auckland, NZ.	Feb. 2021

Posters

Travelling waves and heteroclinic networks in spatially-extended models of cyclic competition. <i>New Zealand Mathematical Society Colloquium.</i> Victoria University of Wellington, NZ.	Dec. 2023
Switching near heteroclinic networks as a piecewise-smooth dynamical system. <i>Antipodeal Dynamics Workshop.</i> University of Exeter, UK and University of Auckland, NZ.	Oct. 2022

Conferences and workshops attended

SIAM Conference on Applications of Dynamical Systems (<i>Upcoming</i>) Organising a two-part minisymposium on dynamics near homoclinic and heteroclinic structures with Alexander Lohse, Universität Hamburg, Germany.	May 2025
Australia and New Zealand Industrial and Applied Mathematics Conference (<i>Upcoming</i>)	Feb. 2025
Joint Meeting of the NZMS, AMS, and AustMS (<i>Upcoming</i>) I serve as the co-ordinator of the Joint Meeting's Social Committee, a group of ECRs organising local information for conference participants and events for ECRs.	Dec. 2024
Pattern Formation Workshop (<i>Upcoming</i>) Satellite meeting to the Joint Meeting of the NZMS, AMS, and AustMS, organised by Dr. Priya Subramanian.	Dec. 2024
Australia and New Zealand Industrial and Applied Mathematics Conference	Feb. 2024
New Zealand Mathematics Research Institute Summer School on Mathematical Neuroscience	Jan. 2024
Antipodeal Dynamics Workshop, University of Exeter and University of Auckland (Virtual) Co-organiser with Prof. Claire Postlethwaite, and Prof. Peter Ashwin and Muhammed Fadera, University of Exeter, UK.	Dec. 2023
New Zealand Mathematical Society Colloquium	Dec. 2023
New Zealand Mathematics and Statistics Postgraduate Student Conference	Nov. 2023
SIAM Conference on Applications of Dynamical Systems	May 2023
New Zealand Mathematics Research Institute Summer School on Groups, Geometry, and Dynamics	Jan. 2023
Sydney Dynamics Group Workshop: Dynamics in NZ	Nov. 2022
Antipodeal Dynamics Workshop, University of Exeter and University of Auckland (Virtual)	Oct. 2022
Australia and New Zealand Industrial and Applied Mathematics Conference (Virtual)	Feb. 2022
Antipodeal Dynamics Workshop, University of Exeter and University of Auckland (Virtual)	Oct. 2021
SIAM Conference on Applications of Dynamical Systems (Virtual)	May 2021
Australia and New Zealand Industrial and Applied Mathematics Conference (Virtual)	Feb. 2021

Professional affiliations

New Zealand Mathematical Society
Australia New Zealand Industrial and Applied Mathematics Society for Industrial and Applied Mathematics
Student member

Employment history

Department of Mathematics, University of Auckland. Research Assistant	Nov. 2021 – Mar. 2022
Suez Smart Solutions Ltd. Assistant Optimisation Engineer	Nov. 2019 – Mar. 2020
Watercare Services Ltd. Undergraduate Engineer	Oct. 2018 – Nov. 2019